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## **5aMUa1. Real-time magnetic resonance imaging of the upper airways during harmonica pitch bends**

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Skilled harmonica players learn to bend the pitch of certain notes by a semitone or more, especially in blues playing, by adjusting the shape of their vocal tract [Bahnon, et al., J. Acoust. Soc. 103, 2134 (2008)]. The changes of the vocal tract have been partially viewed with endoscopy and ultrasound but are still incompletely understood. While in a magnetic resonance imaging (MRI) scanner, a professional harmonica player using nonmagnetic, MRI-compatible diatonic harmonicas played draw and blow notes in both unbent and bent positions. Three-dimensional static and 2-dimensional real-time magnetic resonance images of the upper airway were recorded in the sagittal and coronal planes. We identified and characterized the static and dynamic changes that facilitated pitch bends for low and high notes with specific attention to tongue positioning, tongue morphology, and airway shape. Deliberate changes in the tongue shape are often accompanied by changes in other parts of the vocal tract such as the pharynx.

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## INTRODUCTION

Pitch bending, lowering pitch through changes in the vocal tract, imparts a characteristic and essential feel to harmonica music, especially in blues playing. Skilled harmonica players learn to bend the pitch of certain notes by a semitone or more by adjusting the shape of their vocal tract. This is a challenging technique. Several investigators have described the movement of the harmonica reeds during bending. (Bahnson et al, 1998, Antaki JF et al, 2011, Brock and Cottingham, 2011) while others have discussed mechanisms and proposed models of the changes in the vocal tract responsible for bending (Johnston, R.B., 1987, Millot, L. et al, 2001 and 2011, Cottingham J., 2011, Levy H., 2011). However, the actual configurations of the vocal tract that will produce pitch bending are incompletely understood. There have been only a small number of studies attempting to image the vocal tract during harmonica playing. These studies have been presented on web sites. Ataki J.F., et al (2012) presented studies on harmonica playing including brief x-ray fluoroscopy and ultrasound imaging through the cheek. Bahnson H.T. (2012), using an endoscope in the oral cavity, recorded live viewing of the tongue and larynx during harmonica playing.

The purpose of this paper is to describe the changes that occurred in the vocal tract while a professional harmonica player played different notes as demonstrated with state-of-the-art, magnetic resonance imaging (MRI).

## MATERIALS AND METHODS

### MRI Compatible Harmonicas

Special harmonicas were necessary as no magnetic materials are permissible in the MRI scanner, and even the amount of non-magnetic metal must be minimized to limit circular electrical currents that can produce artifacts in the images. Two MRI-compatible harmonicas were made using a traditional C major 10 hole diatonic harmonica platform. The elements were all plastic except for aluminum mounting bolts, reeds and reed plates. The reed plates were limited in size so they covered only one hole: the third hole on one harmonica to be used for low draw notes and the eighth whole on the second harmonica to be used for higher blow notes.

### MRI Technique

All images were acquired on a 3 Tesla (MR750) MRI scanner (GE Healthcare, Waukesha, WI). A volumetric (3-Dimensional) acquisition was obtained in either the sagittal plane (a vertical plane that divides the body into left and right sections) or coronal plane (a vertical plane that divides the body in to anterior (front) and posterior (back) sections). The sagittal images consist of a “stack” of 32 images with each image a 4mm thick slice of the airway spanning a 32cm x 32 cm field of view. The coronal images also consist of a “stack” of 32 images over a 32cm x 32 cm field of view at a 5mm slice thickness.

Real-time images were also obtained using RTHawk, a flexible, real-time imaging platform (HeartVista, Palo Alto, CA). Dynamic, 2D protocols included synchronous acquisitions in the mid-sagittal plane and coronal planes or mid-sagittal plane only; dynamic acquisitions were obtained at a rate of approximately three images every 2 seconds (Santos J, 2004). Only the static images will be presented in detail in this paper.

### Harmonica Playing and Image Collection

A skilled professional harmonica player and instructor (DB) lay on his back in the MRI scanner. He was able to hold the harmonica normally. Although the MRI scanner produces loud noises during the

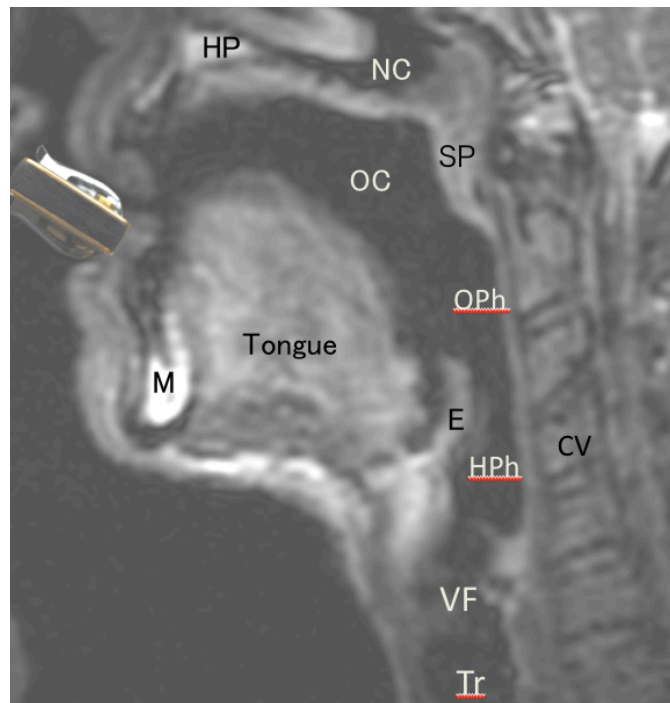
image acquisition process, he could hear the sounds he played and was confident the pitches played were accurate and of good tonal quality despite wearing protective ear plugs.

We collected images of 3 hole draw (inhaling) notes with and without bends and 8 hole blow (exhaling) notes with and without bends. Each note was held constant for the duration of image capture--about 25 seconds. During this time 32 parallel sections were taken through the vocal tract. Both the tongue blocking and pucker or lip blocking embouchure were used. In the tongue blocking technique, the opening of the lips commonly covers 4 holes and the tongue touches and blocks the 3 left holes; in pucker or lip blocking the opening of the lips covers only one hole and the tongue does not touch the harmonica.

## RESULTS

### 3 Hole Draw

A 3 hole draw note with a C major harmonica is B4; with bending, the pitch progressively lowers a semitone, whole tone and or minor third. Images of the playing of each of these 4 notes will be shown. Figure 1 shows the vocal tract during the play of an unbent 3 hole draw note with a tongue block embouchure. It is an image of the sagittal plane near the midline with a representation of the position of the harmonica and labeling of the normal anatomy. The body of the tongue is relatively flat compared to the position in bending, but can show considerable variation without changing the pitch.



**FIGURE 1.** 3 hole draw, no bend, tongue block embouchure with normal MRI anatomy in a representative sagittal MRI image. The drawing shows the position of the harmonica between the lips. Note the body of the tongue (T), nasal cavity (NC), oral cavity or mouth, hard palate (HP), soft palate (SP) which is in a raised position closing the entrance to the nasal cavity, epiglottis (E), cervical vertebrae (CV), mandible or lower jaw (M), vocal folds (VF), and trachea (TR).

When playing the 3 hole draw bent notes, the body of the tongue elevates and humps up toward the hard palate. As the bend deepens, the tongue retracts and the apex of the elevation shifts more and more back toward the pharynx. This process creates a progressively larger cavity in the anterior part of the mouth between the elevated body of the tongue and the lips (Figure 2). Simultaneously, the width of the pharynx

decreases. In these sagittal images, the apposition of the tongue and harmonica is not always seen, because the tongue touches and blocks the holes only the left side of lips. As the plane of section moves through the mouth from the left toward the right side of the mouth the tongue no longer touches the harmonica, exposing the hole being played. The best image to show the shape of the tongue is near the junction of covered and uncovered holes. Looking at multiple planes also reveals that the constricted tube between the tongue and palate is in the midline but neither exactly centered nor exactly straight.

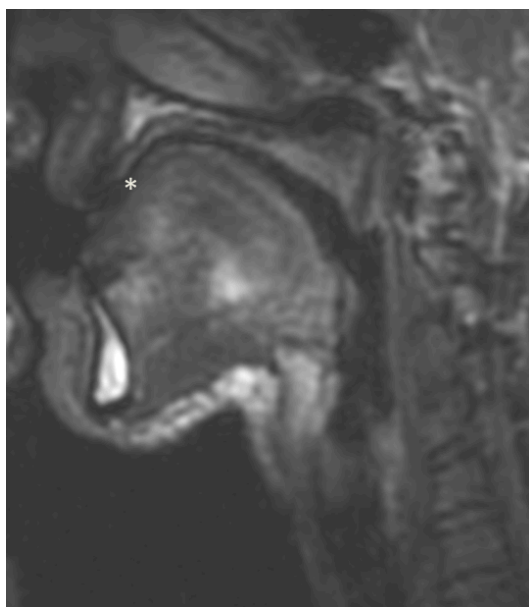


Figure 2A

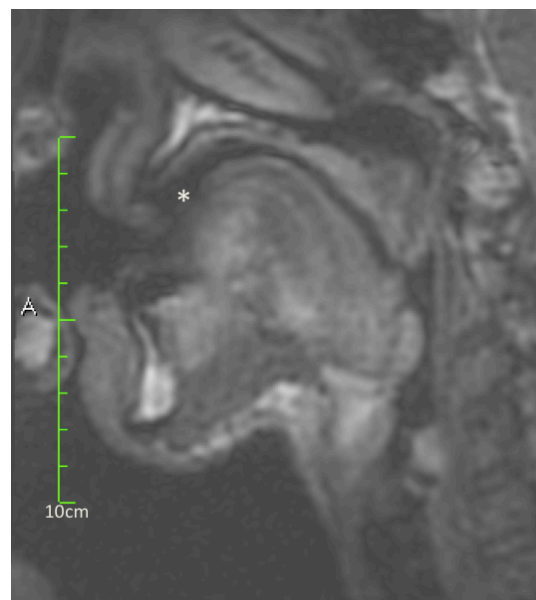


Figure 2B

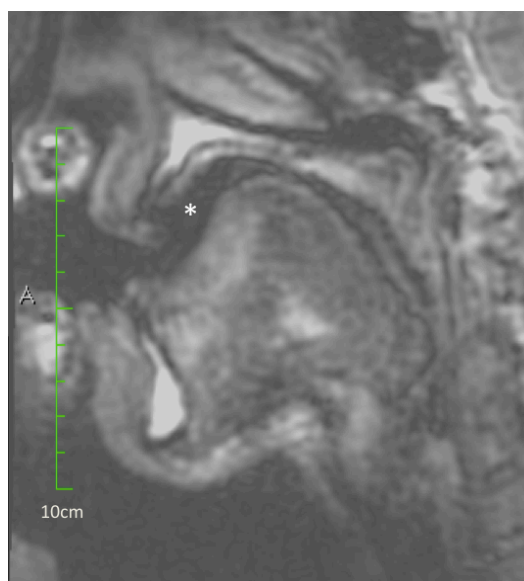
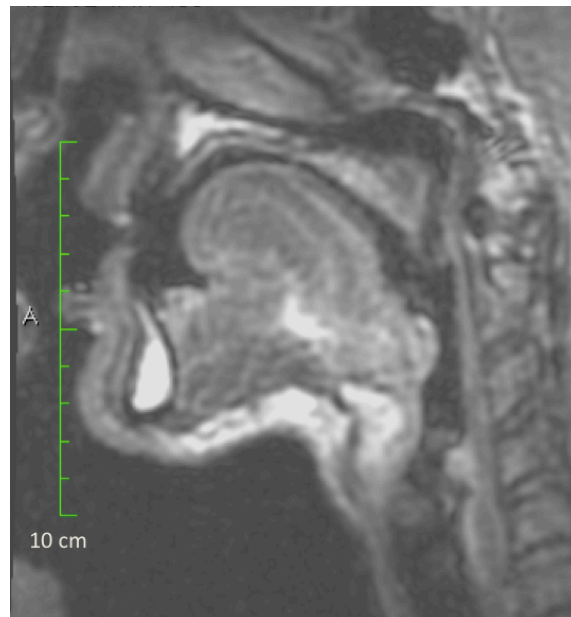


Figure 2C

**FIGURE 2.** 3 hole draw with bends in tongue block embouchure. Sagittal plane near the midline. These images show increasing depths of bending. In all the 3 bends, the body of the tongue is markedly elevated toward the hard palate. The anterior part of the oral cavity—that portion between the elevated mass of the tongue and the harmonica—progressively enlarges. **A.** Half-step bend (pitch B $\flat$ 4). The star indicates the anterior oral cavity. **B.** Whole step bend (pitch A4). This section is just to the right of the area where the tongue touches the harmonica. The star indicates the anterior oral cavity. The green measure line is 10cm with 1cm divisions. **C.** 3 Minor third bend. (pitch A $\flat$ 4). The star indicates the anterior oral cavity. The green measure line is 10cm with 1cm divisions.

### 3 Hole Draw with Pucker Embouchure

The alternate pucker 3 draw whole tone bend embouchure has very similar changes in the elevation and retraction of the body of the tongue. It differs from the tongue blocking technique in that the tip of the tongue is centered and never in contact with the harmonica (Figure 3). Compared to the same 3 draw whole tone bend with tongue blocking (Figure 2B) the tip of the tongue is retracted, yet the position of maximum elevation of the tongue is similar.



**FIGURE 3.** 3 draw whole step bend in pucker or lip block embouchure. The tip of the tongue is retracted away from the harmonica.

Coronal sections give further information for the position of the tongue. In the tongue block embouchure the anterior portion of the tongue is clearly to the left of center (Figure 4A). This is because it is touching the harmonica to block the holes to the left of the one being played. Farther back in the mouth, as the plane of the coronal cuts moves back away from the lips, the tongue is no longer displaced to the left but is centered. It touches the palate except for a small “V” shaped depression in the superior surface of the tongue that, with the palate, forms a narrow channel (Figure 4B). A comparison of all the sequential coronal sections reveals that the ends of the channel flair at both ends. The narrowest constriction in this channel measures approximately  $0.8\text{cm}^2$  in area. The tongue block embouchure produces more space to the right of the tongue than the left so the anterior oral cavity has an asymmetrical shape which is different from that produced by the pucker embouchure. However, we cannot say whether the volumes are different or not.

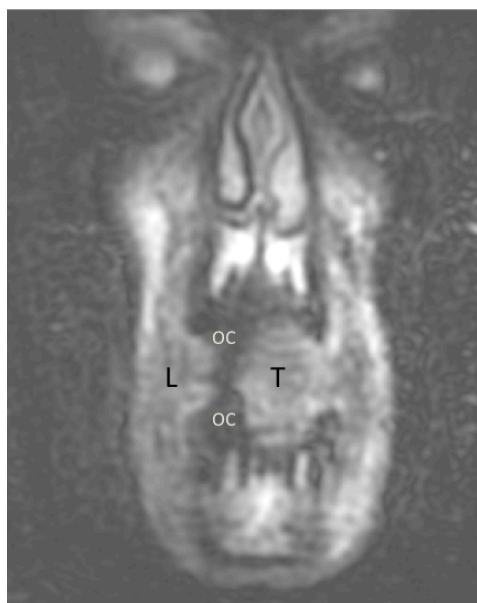


Figure. 4A

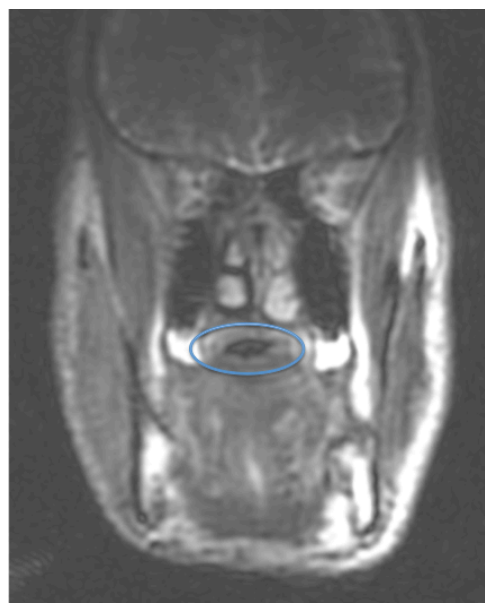


Figure 4B

**FIGURE 4.** 3 draw half step bend with tongue block embouchure. Coronal sections, i.e. images made in planes perpendicular to the long axis of the tongue. **A.** This section is in the anterior mouth, not far from the harmonica. The tongue (T) is off center to the left side typical of the tongue block embouchure. Air is visible in the oral cavity (OC) between and around the tongue and the lips (L) of the right side of mouth. **B.** This section was taken farther back in the mouth, approximately 7cm farther away from the lips than the section shown in Figure 4A. The tongue is centered and touches the palate and teeth except for a narrow center channel (inside the blue oval). The cross sectional area of the channel here is approximately  $0.8 \times 0.8 \text{ cm}^2$ .

## 8 Hole Blow

The 8 hole blow note with a C major harmonica is E5. It can be lowered only one semitone by bending. The sagittal images of an unbent note and a bent note in pucker embouchure are seen in Figure 5. The tongue position in the unbent note can vary somewhat without changing the pitch, although the quality of the tone may vary.

In the bent note, the body of the tongue is elevated toward the palate and the tongue is very far forward next to the lips and harmonica. The anterior surface of the tongue apposes the palate except in the center, leaving a small central channel just as in the 3 hole draw notes. The anterior oral cavity is small. The upward and forward movement of the tongue is more extreme than that found in the images of the 3 hole draw notes, and the pharynx is correspondingly wider.



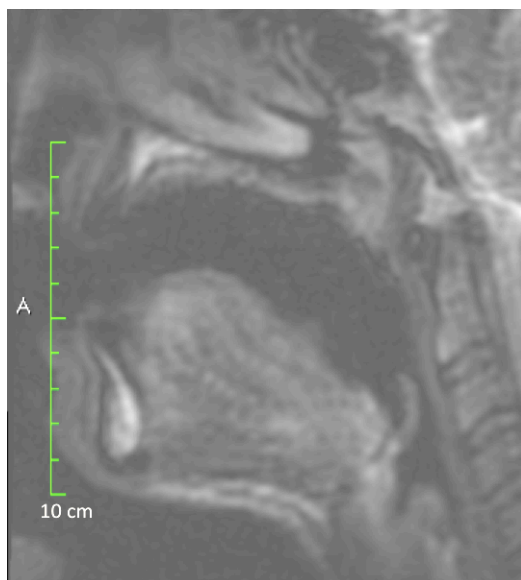


Figure 5A

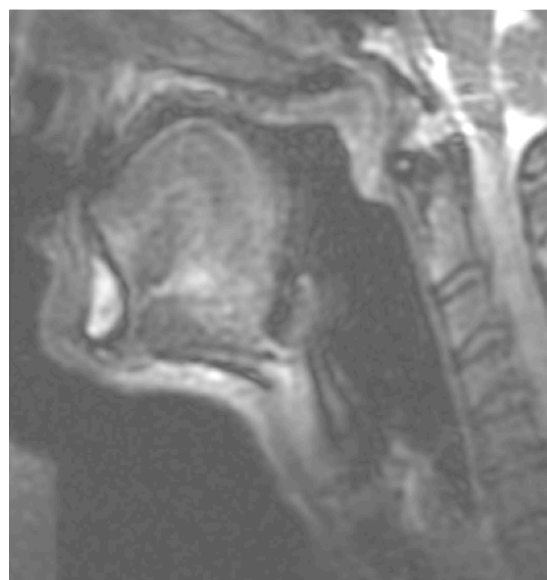


Figure 5B

**FIGURE 5.** 8 hole blow notes with pucker embouchure. Sagittal plane near the midline. **A.** Unbent note (pitch B5). **B.** Half-step bend (pitch Bb5). The body of the tongue is elevated toward the palate and the tip of the tongue is very far forward next to the lips and harmonica.

Real-time dynamic images of the bending process were also obtained. They corroborated the above results from static images and will be presented in a subsequent paper.

## DISCUSSION

This study obtained images that demonstrate many of the anatomical features that are used in pitch bending with the diatonic harmonica. MRI has better soft-tissue contrast and is not as limited at soft-tissue/air interfaces compared with ultrasound, and is superior to x-ray fluoroscopy or computed tomography (CT) scans because no harmful ionizing radiation is used. MR technology has advanced in terms of the time necessary to collect data to generate an image. The MR program used in this study produced 32 parallel images across the vocal tract during a few seconds of continuous playing a single note. The combined images allow a good understanding of 3 dimensional structures of the vocal tract which a single section cannot provide. The use of a coronal plane in addition to the sagittal plane is also very informative. However, the resolution of rapid imaging is limited by lower spatial resolutions acquired at faster temporal resolutions. In addition, there is similar MRI signal between certain soft tissues (e.g. tongue and soft palate) which can limit precise delineation of tissue-tissue boundaries.

The images indicate that the vocal tract during harmonica bending can be divided into 3 parts: an anterior cavity, a posterior cavity and a connecting tube. This is consistent with the theoretical concepts of Johnston (1987) and Millot (2001). The most active element is the tongue. The body of the tongue is elevated such that much of the upper surface touches the hard palate except for a “V” shaped central groove. This groove, along with the mucosa of the palate, forms a tube with flared ends. As the elevated portion of the tongue moves forward or back the size of the anterior and posterior cavities change.

Our MRI planes give a clearer view of the anterior cavity than the posterior one. The coronal plane does not define the largely vertical tissue borders of the pharynx. An understanding of the volume of the posterior cavity is more complicated than the anterior cavity. It changes not only by movement of the tongue, but also by considerable contraction of the muscles of the oropharynx and hypopharynx as well as elevation of the larynx. Since the glottis is open in most of time in harmonica playing, the trachea may also contribute to the volume of the posterior chamber (Johnston, 1987).



The size of the anterior cavity and the position of the tongue show a set of consistent changes which may be very important in the production of pitch bending. The bends on higher notes are associated with a smaller anterior chamber than the bends on the lower notes. The highest of the bent pitches studied, a one half step bend on the 8 hole blow, is associated with an extreme anterior position of the tongue and the smallest anterior cavity. The half step bend on the 3 hole draw is associated with a more posterior tongue position and a larger, although still small, anterior cavity. In deeper bends on this hole the tongue continues to move progressively back and the anterior cavity simultaneously enlarges.

## ACKNOWLEDGEMENT

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